MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, LIGHT EXETNSION

(A) 25 WATTS, 120 VOLTS, EXPLOSIONPROOF AND WATERTIGHT, SYMBOLS 290 AND 291

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

(A)

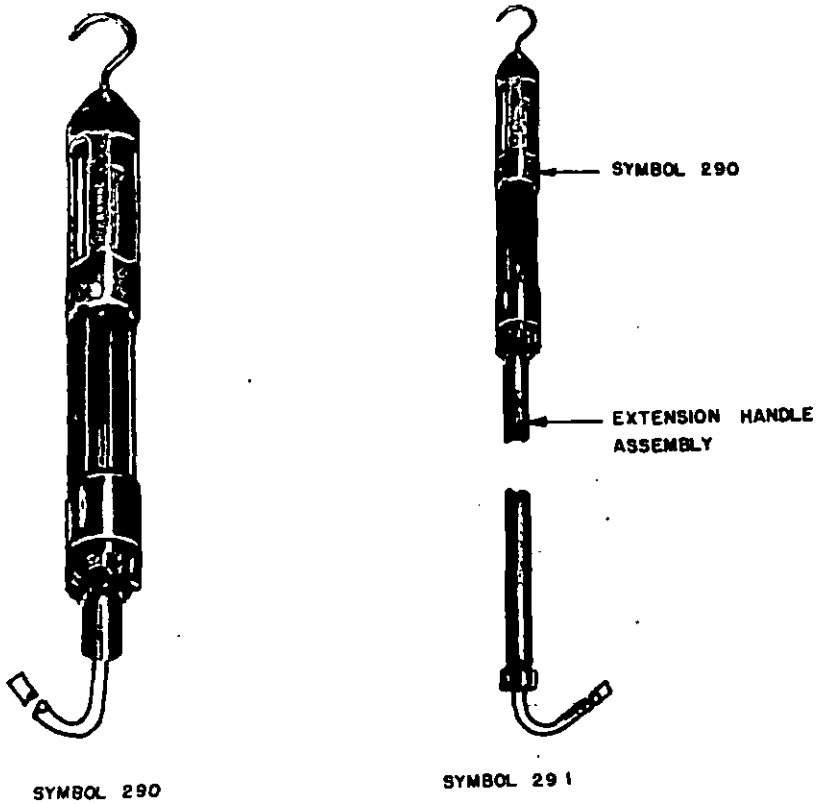


FIGURE 1. Typical view of Symbols 290 and 291.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

(A) Denotes changes

FSC 6230

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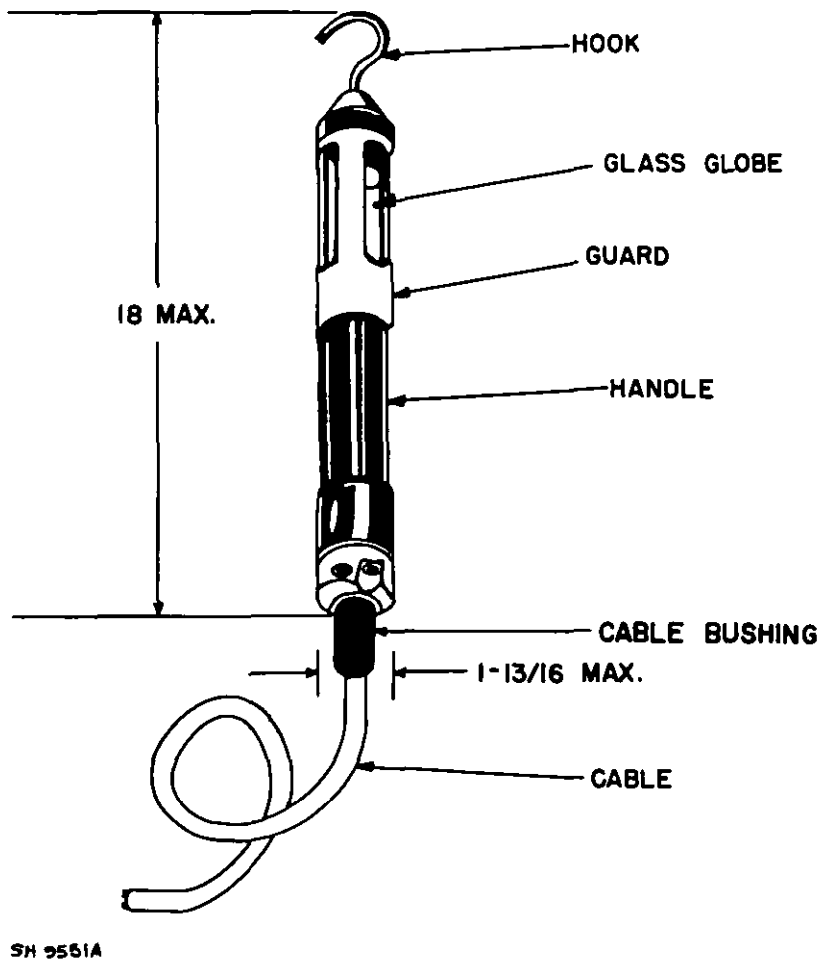


FIGURE 2. Typical view of symbol 290.

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REQUIREMENTS:

Type:	II
Class:	1
Symbol 290:	Light, extension, incandescent, 25 watt, 120 volt explosionproof and watertight.
Symbol 291:	Light, extension, incandescent, 25 watt, 120 volt explosionproof and watertight with extension handle assembly.
Design:	
Electrical rating:	25 watt, 120 volt.
Dimensions and configurations:	See figures 1 and 2.
Metals:	All exposed metal parts shall be made of non-sparking metal.
Guard:	The guard shall be cast or fabricated of nonsparking metal with at least three ribs. One end shall be internally threaded to receive the handle and the other end shall accommodate the hook.
Hook:	The hook shall be located on the outer end of the guard. The design of the hook shall be such that it will fit over 3/4 inch pipe and shall not extend outside of the projected cylindrical surface of the light.
Dimensions:	The overall length from end of hook to cap shall not exceed 18 inches. The extreme outside diameter of the light shall not exceed 1-13/16 inches.
(A) Extension handle assembly:	Extension handle assembly shall be as shown on figure 1.
Handle:	The handle shall be made of nonsparking aluminum or of a high grade plastic compound conforming to MIL-M-14. One end shall be externally threaded to receive the guard and the other end shall accommodate the cable and the extension handle.
Globe:	The globe shall be made of high-impact, heat-resistant, clear glass. The globe shall conform to the requirements specified herein and shall have 85-percent minimum light transmittance.
Lampholder:	The lampholder shall be capable of being easily removed for replacement and shall consist of a spring-loaded center contact to prevent loosening of the lamp. Plastic material shall be of a high-temperature type which will not crack or deteriorate in any manner under normal use. The lampholder assembly shall accommodate the lamp specified herein.
Lamp:	The lamp shall be incandescent, 25-watt, 120-volt, T-6-1/2, intermediate screw, inside frosted, rough service, NSN6240-00295-2005. The lamp shall be furnished with the light.
Enclosure:	The light shall be designed to be explosionproof and watertight and shall be so constructed that no parts will work loose in service. The enclosure shall be of substantial construction and shall be capable of withstanding the internal hydrostatic pressures resulting from explosions without bursting or loosening at the joints. The enclosure shall withstand without rupture or permanent distortion an internal hydrostatic test pressure of four times the maximum internal explosion pressure developed during the explosionproof test. The hydrostatic pressure shall be applied at a rate of about 100 pounds per square inch until the required internal pressure is reached and it shall be held to this pressure for 1 minute.
Cable:	The light shall be furnished with 100 feet (plus 6, minus 0 inches) of 3-conductor cable, type SO-16/3 or ST-16/3 in accordance with J-C-580. The third conductor shall be used for grounding.
Cable pull:	There shall be no strain at the cable connections or damage to the cable when a direct pull of 150 pounds is applied to the cable for one minute. The light shall be anchored by the hook.
Ambient temperature:	The light shall be capable of operating in an ambient temperature of 205°F. for 96 hours without damage or loosening of parts.

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REQUIREMENTS: (continued)

Rough usage: The light shall be capable of withstanding the following rough usage test. The light shall be suspended in a vertical position, with cable entrance up, by means of a short loop formed from a heavy solid wire. The cable shall be removed from the light and one end of the wire loop shall be connected to the light at its cable entrance. The other end of the wire loop shall be extended through a screweye secured to a solidly mounted vertical board, 2 inches thick. The free end of the light shall be pulled away from the board and then allowed to swing back against the board, or a wooden block secured thereto, for 9000 times at a rate of about 50 times per minute, the horizontal swing distance at the threaded engagement for the lamp replacement being about 6 inches. The enclosure and the cable entrance shall not be impaired nor shall threaded engagements be loosened during this test.

Drop: The enclosure and the cable entrance shall not be impaired nor shall threaded engagements be loosened when the light is dropped to a concrete floor ten times from a height of 3 feet. For the first five drops during the test, the light, with cable connected as in service, shall be allowed to freely fall in a manner that would normally be the case if the device were to fall from a horizontal platform to the floor. In the remainder of the test, the light with cable shall be held at various angles about 3 feet above the concrete floor and then dropped to the floor.

Weight:

Symbol 290: 2 pounds maximum without cable; 10-1/2 pounds with cable.

Extension handle assembly: 18 ounces maximum.

Certification of approval by Underwriters' Laboratories, Incorporated: Prior to submission of lights for acceptance, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that Underwriters' Laboratories, Incorporated has approved the item as an electric lighting fixture suitable for use in class I, group D hazardous locations in accordance with UL-781. In the event of any conflict between UL-781 and this specification, the latter shall govern.

Identification and information plates: Identification and information plates shall be in accordance with MIL-P-16377. The following plates shall be furnished:

(a) Identification plate. The following information shall appear on an identification plate securely attached to the housing:

Light, extension, explosionproof, and watertight
MIL-P-16377/51 dated
Symbol 290
NSN6230-00-283-9671
Contract No. (To be supplied by the procuring activity)
Manufacturer's name or trademark

(b) Warning plate. A warning plate bearing the following legend, in upper case letters, shall be provided:

"USE LAMP NSN6240-00-295-2005"

"FOR USE IN THE FOLLOWING EXPLOSIVE ATMOSPHERES ONLY: GASOLINE, PETROLEUM, NAPHTHA, ALCOHOL, ACETONE, LACQUER SOLVENT VAPORS, AND NATURAL GAS."

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REQUIREMENTS: (continued)

CAUTION

Disconnect from circuit before relamping. Keep all joints tight when in operation. If the connection to the electrical circuit is in an area where explosive gases are present, an explosionproof plug and matching receptacle shall be used. If the electrical connection is made outside of the hazardous area, a 3-conductor grounded plug may be used. The lamps shall not be replaced or the light disassembled in a hazardous location.

(c) Underwriters' Laboratories label plate. A label plate shall be provided in accordance with Underwriters' Laboratories requirements.

Quality assurance: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection: See table I.
Quality conformance:
Comparison inspection: Same as the first article inspection.
Inspection of product for delivery:
Examination: See MIL-F-16377
Examination for preparation of delivery: See MIL-F-16377
Quality conformance tests: See MIL-F-16377

TABLE I. First article and comparison inspection.

Inspection	Requirement paragraph	Inspection paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Rough usage	1/	1/	
Drop	1/	1/	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Ambient temperature	1/	1/	
Grounding circuit	3.6.13.3	4.8.11	
Leakage current	3.6.13.5	4.8.13	
Optical uniformity	3.7.5.1	4.8.4	
Thermal shock	3.7.5.7	4.8.17.4	
Color and luminous transmittance	3.7.3.1	4.8.6.3	
Magnetic permeability	3.4.7	4.8.18	
Cable pull	1/	1/	
Salt spray	3.5.15	4.8.10	
Enclosure effectiveness	3.5.11	4.8.15	Explosion-proof
Hydrostatic pressure	1/	1/	

1/ As specified in this specification.

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General information: Assignments of Military part numbers and general information is as follows:

(a) Military part number:
Light-Symbol 290: M16377/51-290
Light-Symbol 291: M16377/51-291
Extension handle: M16377/51-001

(b) National Stock Number (NSN):
Light-symbol 290: 6230-00-283-9671
Extension handle: 6230-00-299-7052

(c) The light specified in this specification was previously covered by MIL-L-24349 and MIL-L-21751 which have now been superseded by this specification.

(d) Light symbol 290 is intended to be used as a general purpose extension light for detail illumination onboard ship and as a work light in areas where explosive gases may be present. When the extension handle is assembled to this light, the light as such (symbol 291) is intended to be used for examining the interior of airplane wiring tanks, barrels, and drums.

Preparing activity:
Navy - SH
(Project 6230- N280)

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